

AR-Based Real-Time Speech Transcription Glasses for the Hearing Impaired

Introduction

Communication is a fundamental human necessity that enables social participation, education, and professional growth. For individuals who are deaf or hard of hearing (DHoH), interaction with the hearing population remains challenging due to the limited adoption of sign language among the general public. This communication divide often leads to social exclusion, reduced independence, and reliance on intermediaries. Advances in Automatic Speech Recognition (ASR), embedded computing, and augmented reality (AR) have created opportunities to design wearable assistive technologies that are both effective and socially inclusive. This project proposes an AR-based Head-Up Display (HUD) smart glasses system that provides real-time speech-to-text transcription directly within the user's field of vision. By projecting text onto a transparent optical display, the system enables hands-free communication while allowing users to maintain eye contact and perceive essential non-verbal cues such as facial expressions and gestures.

Problem Statement

Deaf and hard-of-hearing individuals frequently experience communication barriers in everyday interactions, educational environments, and workplaces. Although smartphone-based speech transcription applications are widely available, they require users to divert their gaze away from the speaker to read the screen. This disrupts natural conversation and causes loss of contextual information conveyed through body language and facial expressions. Additionally, handheld devices are inconvenient in situations that demand mobility, multitasking, or professional etiquette. Environmental noise, varying accents, and speaker distance further reduce transcription accuracy and usability.

Objectives

The primary objectives of this project are to design and develop a lightweight wearable HUD-based smart glasses system for real-time speech transcription, implement a high-accuracy and low-latency speech-to-text pipeline with an end-to-end delay of less than 500 milliseconds, utilize a Raspberry Pi-based

processing engine for audio and system control, integrate a noise-resilient audio acquisition system, and ensure ergonomic comfort and portability for extended daily use.

Methodology

Speech is captured using a high-sensitivity MEMS microphone mounted on the glasses frame. A Raspberry Pi Zero 2W serves as the central processing unit, performing audio preprocessing, system control, and connectivity management. The processed audio stream is forwarded to a speech-to-text engine such as OpenAI Whisper or Vosk. The resulting transcription is formatted for readability and projected onto a transparent optical combiner using a micro-OLED or LCoS display, ensuring a comfortable focal distance for the user.

Performance Measures

System performance is evaluated using word error rate for transcription accuracy, end-to-end latency from speech input to display output, battery life with a target of 4–6 hours, total device weight and comfort, and robustness across varying noise and lighting conditions.

Hardware Requirements

The hardware setup includes a Raspberry Pi Zero 2W, a micro-OLED or LCoS display, a beam splitter or prism-based optical combiner, a high-sensitivity MEMS microphone, integrated Wi-Fi and Bluetooth connectivity, a Li-Po battery with charging circuitry, and a custom 3D-printed eyewear enclosure.

Software Requirements

The software stack consists of Raspberry Pi OS Lite, Python and C++ for development, OpenAI Whisper or Vosk for speech recognition, OpenCV or LVGL for text rendering, and WebSockets or MQTT for real-time data communication.

Expected Outcomes

The expected outcome is a functional AR-based smart glasses prototype capable of real-time speech transcription with minimal latency, improved conversational flow for DHoH users, and effective operation in moderately noisy environments.

Evaluation and Assessment

Evaluation includes technical testing for transcription accuracy, latency analysis, user acceptance testing for comfort and readability, and benchmarking against existing smartphone-based transcription solutions.

Scalability and Future Development

Future enhancements include integrating a camera for sign language recognition using computer vision, adding obstacle detection capabilities to assist visually impaired users, developing a lightweight smart glasses operating system supporting downloadable applications such as navigation and widgets, and incorporating a multimodal AI assistant capable of interacting with the user based on audio and visual inputs.

Conclusion

The proposed AR-based real-time speech transcription glasses present an inclusive and scalable assistive technology. By combining Raspberry Pi-based edge computing, advanced speech recognition, and optical HUD systems, the solution enables natural, hands-free communication for hearing-impaired users while offering a strong foundation for future multimodal assistive applications.